Subject: OPTIMIZATION TECHNIQUES

Subject Code: 23AS2402e

UNIT I : Introduction and Linear Programming

Syllabus: Meaning, nature, scope and significance of optimization, typical applications; formulation of LPP, limitations, graphical method, simplex method (maximization and minimization), Big-M method and Two-Phase method.

I. Introduction to Optimization

Definition:

Optimization is the process of finding the best possible solution (maximum or minimum) of an objective, from all feasible alternatives, subject to given restrictions (constraints).

- Objective: maximise profit, revenue, efficiency; minimise cost, time, waste, risk.
- Origin: Operations Research (O.R.) developed in World War II (UK, 1940) for scarce military resources.
- O.R. = scientific, quantitative approach to decision making; uses mathematical models.

Nature of optimization

- Has a decision (controllable) variable set, an objective function, and constraints.
- Mathematical model: Optimise $Z = f(x1, \ldots, xn)$ subject to $g_i(x) \leq , = , \geq b_i$.
- Deals with limited resources: men, machines, materials, money, time.
- Interdisciplinary: mathematics, economics, engineering, statistics, computers.

Scope and significance

- Scope: production planning, inventory, finance and investment, marketing, transport and logistics, personnel, project planning, agriculture, defence, health care.
- Gives scientific basis for decisions instead of guess work; saves cost and time.
- Helps in resource allocation, co-ordination of departments and long term planning.

Typical applications

Area	Application
Production	Product mix, blending, job sequencing
Transport	Minimum cost shipping, vehicle routes
Finance	Portfolio selection, capital budgeting
Marketing	Media selection, advertising budget
Personnel	Assignment of staff, shift scheduling
Projects	CPM / PERT scheduling

Phases of an O.R. study

- 1) Define the problem 2) Construct the model 3) Solve the model 4) Validate the model 5) Implement the solution.

2. Linear Programming Problem (LPP) – Formulation

Definition:

Linear Programming is a technique for optimising a ***linear*** objective function subject to ***linear*** equality / inequality constraints, with non-negative variables.

General mathematical form

Formula:

Max (or Min) $Z = c_1x_1 + c_2x_2 + \dots + c_nx_n$ (objective function)

subject to $a_{i1}x_1 + a_{i2}x_2 + \dots + a_{in}x_n$ ($\leq$, $=$, $\geq$) b_i, $i = 1, \dots, m$ (constraints)

$x_1, x_2, \dots, x_n \geq 0$ (non-negativity restriction)

- Decision variables x_j; cost / profit coefficients c_j; technological coefficients a_{ij}; right hand side (resources) b_i.
- Feasible solution: satisfies all constraints. Feasible region: set of all feasible solutions.
- Optimal solution: feasible solution that gives the best value of Z. Basic feasible solution: at most m non-zero variables.

Assumptions of LPP

- Linearity (proportionality and additivity), certainty (constants known), divisibility (fractions allowed), non-negativity, single objective.

Steps in formulation

- 1) Identify the decision variables. 2) Write the objective function. 3) Write constraints from resources. 4) Add $x \geq 0$.

Limitations of LPP

- Needs linear relations – real life is often non-linear.
- Assumes certainty of coefficients; no uncertainty / probability.
- Answers may be fractional (integer programming needed for whole numbers).
- Only one objective (goal programming for many); large problems need computers.

Example:

Product mix: A firm makes tables (profit Rs 40) and chairs (Rs 30). Each table needs 2 h of carpentry and each chair 1 h; 16 h are available. Total production capacity is 12 units. Formulate.

$x1$ = tables, $x2$ = chairs. Max $Z = 40x1 + 30x2$

s.t. $2x1 + x2 \leq 16$ (carpentry), $x1 + x2 \leq 12$ (capacity), $x1, x2 \geq 0$

Example:

Diet problem: A diet needs at least 18 units of vitamin V1 and 12 units of V2. Food 1 gives (6, 1) units per kg at Rs 20/kg; food 2 gives (1, 4) units per kg at Rs 40/kg. Formulate.

Min $Z = 20x1 + 40x2$ s.t. $6x1 + x2 \geq 18$, $x1 + 4x2 \geq 12$, $x1, x2 \geq 0$

Example:

Blending: Mix two ores to make at least 30 kg of alloy with at most 40 % impurity. Let $x1$, $x2$ be kg used: $x1 + x2 \geq 30$; $0.5x1 + 0.3x2 \leq 0.4(x1 + x2)$ $\Rightarrow 0.1x1 - 0.1x2 \leq 0$.

Note:

Check units of every constraint and make sure each variable appears in the objective.

3. Graphical Method (two variables)

- Step 1: Convert each inequality to an equation and draw the line.
- Step 2: Mark the side that satisfies the inequality; common area = feasible region.
- Step 3: Find the corner (extreme) points of the feasible region.

- Step 4: Evaluate Z at every corner point. Step 5: Max → largest Z, Min → smallest Z.

Note:

Fundamental theorem: if an optimal solution exists, it occurs at a corner point of the (convex) feasible region.

Example:

Solve: Max Z = 40x1 + 30x2 s.t. x1 + x2 ≤ 12, 2x1 + x2 ≤ 16, x1, x2 ≥ 0.

- Lines: x1 + x2 = 12 meets axes at (12, 0), (0, 12); 2x1 + x2 = 16 meets axes at (8, 0), (0, 16).
- Intersection: subtract the equations ⇒ x1 = 4, x2 = 8.

Corner point	Z = 40x1 + 30x2
O (0, 0)	0
A (8, 0)	320
B (4, 8)	400
C (0, 12)	360

- ***Optimal: x1 = 4, x2 = 8, Max Z = 400.***

Example:

Solve (diet): Min Z = 20x1 + 40x2 s.t. 6x1 + x2 ≥ 18, x1 + 4x2 ≥ 12, 2x1 + x2 ≥ 10.

- Feasible region is unbounded, lying above all the lines. Corner points: (0, 18), (2, 6), (4, 2), (12, 0).
- Intersections: 6x1 + x2 = 18 and 2x1 + x2 = 10 ⇒ (2, 6); 2x1 + x2 = 10 and x1 + 4x2 = 12 ⇒ (4, 2).

Corner point	Z = 20x1 + 40x2
(0, 18)	720

(2, 6)	280
(4, 2)	160
(12, 0)	240

- **Optimal: x1 = 4, x2 = 2, Min Z = 160.**

Special cases

- **Multiple optimal solutions:** objective line parallel to a binding constraint – every point on that edge is optimal (e.g. Max Z = 4x1 + 2x2 with 2x1 + x2 ≤ 8).
- **Unbounded solution:** feasible region unbounded in the direction of improvement; Z can be increased without limit.
- **Infeasible solution:** no common region (constraints contradict).
- **Redundant constraint:** does not touch the feasible region – can be dropped.

4. Simplex Method – Preliminaries

Definition:

Simplex method (G. B. Dantzig, 1947) is an iterative algebraic procedure that moves from one basic feasible solution (corner) to a better one until optimum is reached.

Standard form – convert all constraints to equations

Constraint	Add	Cost in objective
≤	slack variable s (+s)	0
≥	surplus variable (–s) and artificial A	0 and M (Big-M)
=	artificial variable A (+A)	M

- All b_i must be ≥ 0 (multiply by -1 if necessary, reversing the inequality). Unrestricted variable $x = x' - x''$.

Terms

- ***Basic variables:*** variables in the basis (value from RHS); non-basic variables are zero.
- C_B: cost of basic variables; $Z_j = \Sigma C_B \cdot$ (column j); net evaluation $\Delta_j = C_j - Z_j$.
- ***Key (pivot) column:*** entering variable. ***Key row:*** leaving variable (min ratio). ***Pivot element:*** intersection.

Simplex algorithm

- Step 1: Write the standard form and the initial table (slack variables form the initial basis).
- Step 2: Compute Δ_j. ***Max problem:*** optimal if all $\Delta_j \leq 0$. ***Min problem:*** optimal if all $\Delta_j \geq 0$.
- Step 3: Entering variable = most positive Δ_j (max) / most negative (min).
- Step 4: Ratio = RHS / (positive entries of key column); smallest ratio gives the leaving variable.
- Step 5: New key row = old key row ÷ pivot. Other rows: new row = old row − (its key column entry) × new key row.
- Step 6: Repeat until optimal.

5. Simplex Method – Maximisation Example

Example:

Max $Z = 3x1 + 5x2$ s.t. $x1 \leq 4$, $2x2 \leq 12$, $3x1 + 2x2 \leq 18$, $x1, x2 \geq 0$.

- Standard form: $x1 + s1 = 4$; $2x2 + s2 = 12$; $3x1 + 2x2 + s3 = 18$. Cj = 3, 5, 0, 0, 0.

- **Table 1** (Δj = 3, 5, 0, 0, 0 → x2 enters; ratios 12/2 = 6, 18/2 = 9 → s2 leaves, pivot 2)

Basis	x1	x2	s1	s2	s3	RHS
s1	1	0	1	0	0	4
s2	0	**2**	0	1	0	12
s3	3	2	0	0	1	18
Δj	3	5	0	0	0	$Z = 0$

- **Table 2** (Δj = 3, 0, 0, -2.5, 0 → x1 enters; ratios 4/1 = 4, 6/3 = 2 → s3 leaves, pivot 3)

Basis	x1	x2	s1	s2	s3	RHS
s1	1	0	1	0	0	4
x2	0	1	0	1/2	0	6
s3	**3**	0	0	-1	1	6
Δj	3	0	0	-5/2	0	$Z = 30$

- **Table 3** (Δj = 0, 0, 0, -3/2, -1 → all $\leq 0 \Rightarrow$ **optimal**)

Basis	x1	x2	s1	s2	s3	RHS
s1	0	0	1	1/3	-1/3	2
x2	0	1	0	1/2	0	6
x1	1	0	0	-1/3	1/3	2
Δj	0	0	0	-3/2	-1	$Z = 36$

- **Optimal solution: x1 = 2, x2 = 6, Max Z = 36.**

Note:

All $\Delta j \leq 0$ but one non-basic variable with $\Delta j = 0 \Rightarrow$ alternative optimal solutions exist.

6. Simplex Method – Minimisation Model

- **Method 1:** convert to a maximisation problem: Min $Z = -\text{Max}(-Z)$. Solve Max $Z' = -Z$, then Min $Z = -Z'$.
- **Method 2:** use the same tables but enter the variable with the most negative Δj and stop when all $\Delta j \geq 0$ (Big-M if $\geq$ constraints).

Example:

Min $Z = -2x1 - x2$ s.t. $x1 + x2 \leq 5$, $x1 \leq 3$, $x1, x2 \geq 0$.

- Put $Z' = -Z = 2x1 + x2$ and maximise. Standard: $x1 + x2 + s1 = 5$; $x1 + s2 = 3$.

Basis	x1	x2	s1	s2	RHS
s1	1	1	1	0	5
s2	1	0	0	1	3
Δj	2	1	0	0	$Z' = 0$

- x1 enters, ratios 5 and 3 ⇒ s2 leaves.

Basis	x1	x2	s1	s2	RHS
s1	0	1	1	-1	2
x1	1	0	0	1	3
Δj	0	1	0	-2	$Z' = 6$

- x2 enters; ratio 2 ⇒ s1 leaves.

Basis	x1	x2	s1	s2	RHS
x2	0	1	1	-1	2
x1	1	0	0	1	3
Δj	0	0	-1	-1	$Z' = 8$

- Optimal: $x1 = 3$, $x2 = 2$, $Z' = 8$ ⇒ **Min $Z = -8$.**

Special situations in simplex

- **Unbounded:** entering column has no positive entries (no ratio) ⇒ Z unbounded.
- **Degeneracy:** tie in minimum ratio ⇒ a basic variable becomes 0 (cycling is rare).
- **Infeasible:** optimal table still contains an artificial variable with positive value.
- **Alternative optima:** a non-basic variable has $\Delta j = 0$ at optimum.

7. Big-M Method (Charnes' penalty method)

Definition:

Used when the constraints include ≥ or = (no obvious starting basis). Artificial variables are added with a very large penalty M in the objective (+M for Min, -M for Max).

- Artificial variables are forced out of the basis by the penalty; once an artificial variable leaves, its column is dropped.
- If an artificial variable stays in the final basis at positive level ⇒ the problem has no feasible solution.

Example:

Min Z = 4x1 + x2 s.t. 3x1 + x2 = 3, 4x1 + 3x2 ≥ 6, x1 + 2x2 ≤ 4, x1, x2 ≥ 0.

- Standard form: 3x1 + x2 + A1 = 3; 4x1 + 3x2 − S2 + A2 = 6; x1 + 2x2 + S3 = 4. Cj = 4, 1, 0, 0, M, M.
- Initial basis A1, A2, S3. $\Delta x1 = 4 - 7M$, $\Delta x2 = 1 - 4M$, $\Delta S2 = M$. x1 enters (most negative); ratios 1, 1.5, 4 ⇒ A1 leaves.

Basis	xl	x2	S2	S3	A2	RHS
x1	1	1/3	0	0	0	1

A2	0	5/3	-1	0	1	2
S3	0	5/3	0	1	0	3

- $\Delta x2 = -(1 + 5M)/3$ is most negative: x2 enters; ratios 3, 6/5, 9/5 ⇒ A2 leaves (its column is dropped).

Basis	x1	x2	S2	S3	RHS
x1	1	0	1/5	0	3/5
x2	0	1	-3/5	0	6/5
S3	0	0	1	1	1

- $\Delta S2 = -1/5$ (negative) ⇒ S2 enters; ratios 3, -, 1 ⇒ S3 leaves.

Basis	x1	x2	S2	S3	RHS
x1	1	0	0	-1/5	2/5
x2	0	1	0	3/5	9/5
S2	0	0	1	1	1

- $\Delta S3 = 1/5 \geq 0$ ⇒ **optimal: x1 = 2/5, x2 = 9/5, Min Z = 17/5 = 3.4.**

Note:

Check: $3(0.4) + 1.8 = 3$ ✓; $4(0.4) + 3(1.8) = 7 \geq 6$ ✓; $0.4 + 3.6 = 4$ ✓.

8. Two-Phase Method

Definition:

Avoids the large number M. **Phase I** finds a basic feasible solution; **Phase II** optimises the original objective.

- **Phase I:** replace the objective by W = sum of artificial variables, Min W (cost 0 for all real variables, 1 for each artificial).
  - If Min W = 0 → feasible solution exists, go to Phase II. If Min W > 0 → problem infeasible, stop.
- **Phase II:** take the final Phase I table (drop artificial columns), restore the original Cj, recompute Δj and continue simplex.

Example:

Same problem as before: Min Z = 4x1 + x2 with the three constraints.

- **Phase I:** Min W = A1 + A2. Initial Δx1 = -7, Δx2 = -4, ΔS2 = +1 ⇒ x1 enters, A1 leaves.
- Then Δx2 = -5/3 ⇒ x2 enters, A2 leaves. Table: x1 = 3/5, x2 = 6/5, S3 = 1, both artificials out ⇒ **Min W = 0**.

Basis	x1	x2	S2	S3	RHS
x1	1	0	1/5	0	3/5
x2	0	1	-3/5	0	6/5
S3	0	0	1	1	1

- **Phase II:** Cj = 4, 1, 0, 0. ΔS2 = 0 – (4/5 – 3/5) = -1/5 ⇒ S2 enters, S3 leaves.
- Next table gives ΔS3 = 1/5 ≥ 0 ⇒ optimal: **x1 = 2/5, x2 = 9/5, Z = 17/5** (same answer as Big-M).

Big-M	Two-Phase
One big table, M appears in Δj	Two simple tables, numbers only
Computer round-off with M	No M needed – more stable
Same final answer	Same final answer

9. Practice Problems with Answers

- 1) Max $Z = 5x1 + 4x2$ s.t. $6x1 + 4x2 \leq 24$, $x1 + 2x2 \leq 6$, $x1, x2 \geq 0$ (graphical). Ans: corners (0,0), (4,0), (3, 1.5), (0,3); $Z = 0, 20, 21, 12 \Rightarrow x1 = 3, x2 = 1.5, Z = 21$.
- 2) Max $Z = 2x1 + 3x2$ s.t. $x1 + x2 \leq 4$, $x1 + 3x2 \leq 6$. Ans: corners (0,0), (4,0), (3, 1), (0,2): $Z = 0, 8, 9, 6 \Rightarrow Z = 9$ at (3, 1).
- 3) Min $Z = 2x1 + 3x2$ s.t. $x1 + x2 \geq 5$, $x1 \geq 2$. Ans: corners (2,3), (5,0): $Z = 13, 10 \Rightarrow$ Min $Z = 10$ at (5, 0).
- 4) Why is the simplex ratio test required? Ans: to keep every basic variable ≥ 0 (feasibility).
- 5) Formulate: A firm needs at least 10 units of A and 12 units of B. Machine 1 gives 2A + 3B per hour (cost Rs 5/h), Machine 2 gives 4A + 2B (Rs 4/h).
  - Ans: Min $Z = 5x1 + 4x2$ s.t. $2x1 + 4x2 \geq 10$, $3x1 + 2x2 \geq 12$, $x1, x2 \geq 0$.

> **Note:**
> Duality is excluded from this syllabus – concentrate on tabular method, Big-M and Two-Phase.

Quick Revision – Unit I

- LPP = linear Z, linear constraints, $x \geq 0$. Optimum lies at a corner point.
- $\leq \rightarrow$ slack (cost 0); $\geq \rightarrow$ surplus + artificial; $= \rightarrow$ artificial.
- Max optimal when all $\Delta j \leq 0$; Min optimal when all $\Delta j \geq 0$; $\Delta j = Cj - Zj$.
- Leaving variable = minimum positive ratio (RHS / key column).
- Big-M: artificial cost M. Two-Phase: Phase I Min W (Σ artificial), Phase II original Z.